0062796S9 Colin Morris
Jesse Sida
London Mills
Angel Jurado
Julian Ramos

Worksheet 2

1. Alan leaves Los Angeles at 8:00 AM to drive to San Francisco, 400 mi away. He travels at a steady 50 mph. Beth leaves Los Angeles at 9:00 AM and drives a steady 60 mph

a. Who gets to San Francisco first?
b. How long does the first to arrive have to wait for the second?

a. $d = 400\,mi$
$A = 50 mph$ Alan leaves @ 8:00 AM
$B = 60 mph$ Beth leaves @ 9:00 AM

$A = \frac{50 mi}{h}$

$B = \frac{60 mi}{h}$

t_f of first

$t_{ff}; t_{fs}$

$t_{fs} - t_{ff} = \Delta t_f$

400 mi

LA ——— SF

$A = \frac{50 mi}{h}$

$B = \frac{60 mi}{h}$

$V_{avg} = \frac{\Delta X}{\Delta t}$

$V = \frac{X_f - X_i}{t_f - t_i}$

$V(t_f - t_i) = X_f - X_i$

$t_f - t_i = \frac{X_f - X_i}{V}$

$$t_f = \frac{x_f - x_i}{v} + t_i$$

Solve for Δt or t_i

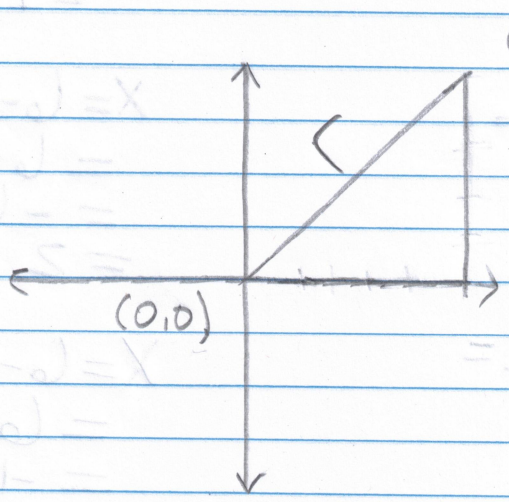
$$t_f = \frac{400\,mi - 0\,mi}{60\,mi/h} + 9AM$$

2.

position v. time
velocity v. time
acceleration v. time

3. $(x,y) = (0,0)$ to $(3,4)$ in 5 seconds

$$\Delta r = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$



$$\Delta r = \sqrt{(3-0)^2+(4-0)^2}$$
$$= \sqrt{3^2+4^2}$$
$$= \sqrt{9+16}$$
$$= \sqrt{25}$$
$$= 5$$

$$V_{avg} = \frac{\Delta r}{\Delta t}$$

$$V_{avg} = \frac{5}{5}$$

$$\boxed{V_{avg} = 1}$$

4. $X = 6 - 6t + t^2$

X measured in meters
t measured in seconds

a. Find the values of for $t = 1, 2, 3, 4, 5$

X(m)	t(s)
1	1
2	2
3	3
4	-2
5	1

$$\frac{dx(t)}{d(t)} =$$

$V(t) = -6 + 2t$

$X = 6 - 6t + t^2$
$= 6 - 6(1) + (1)^2$
$= 0 + 1$
$= 1$

$X = 6 - 6(2) + (2)^2$
$= 6 - 12 + 4$
$= -6 + 4$
$= 2$

$X = 6 - 6(3) + 3^2$
$= 6 - 18 + 9$
$= -12 + 9$
$= 3$

$X = 6 - 6(4) + 4^2$
$= 6 - 24 + 16$
$=$

$X = 6 - 6(5) + 5^2$
$= 6 - 30 + 25$
$=$